

XCS221 Assignment 3 — Route Planning

Due Sunday, September 20 at 11:59pm PT.

Guidelines

1. If you have a question about this homework, we encourage you to post your question on our Slack channel, at <http://xcs221-scpd.slack.com/>
2. Familiarize yourself with the collaboration and honor code policy before starting work.
3. For the coding problems, you must use the packages specified in the provided environment description. Since the autograder uses this environment, we will not be able to grade any submissions which import unexpected libraries.

Submission Instructions

Written Submission: Some questions in this assignment require a written response. For these questions, you should submit a PDF with your solutions online in the online student portal. As long as the PDF is legible and organized, the course staff has no preference between a handwritten and a typeset $\text{\LaTeX}$ submission. If you wish to typeset your submission and are new to $\text{\LaTeX}$, you can get started with the following:

- Type responses only in `submission.tex`.
- Submit the compiled PDF, **not** `submission.tex`.
- Use the commented instructions within the `Makefile` and `README.md` to get started.

Also note that your answers should be in order and clearly and correctly labeled to receive credit. Be sure to submit your final answers as a PDF and **tag all pages correctly when submitting to Gradescope**.

Coding Submission: Some questions in this assignment require a coding response. For these questions, you should submit only the `src/submission.py` file in the online student portal. For further details, see Writing Code and Running the Autograder below.

Honor code

We strongly encourage students to form study groups. Students may discuss and work on homework problems in groups. However, each student must write down the solutions independently, and without referring to written notes from the joint session. In other words, each student must understand the solution well enough in order to reconstruct it by him/herself. In addition, each student should write on the problem set the set of people with whom s/he collaborated. Further, because we occasionally reuse problem set questions from previous years, we expect students not to copy, refer to, or look at the solutions in preparing their answers. It is an honor code violation to intentionally refer to a previous year's solutions. More information regarding the Stanford honor code can be found at <https://communitystandards.stanford.edu/policies-and-guidance/honor-code>.

Writing Code and Running the Autograder

All your code should be entered into `src/submission.py`. When editing `src/submission.py`, please only make changes between the lines containing `### START_CODE_HERE ###` and `### END_CODE_HERE ###`. Do not make changes to files other than `src/submission.py`.

The unit tests in `src/grader.py` (the autograder) will be used to verify a correct submission. Run the autograder locally using the following terminal command within the `src/` subdirectory:

```
$ python grader.py
```

There are two types of unit tests used by the autograder:

- **basic:** These tests are provided to make sure that your inputs and outputs are on the right track, and that the hidden evaluation tests will be able to execute.
- **hidden:** These unit tests are the evaluated elements of the assignment, and run your code with more complex inputs and corner cases. Just because your code passed the basic local tests does not necessarily mean that they will pass all of the hidden tests. These evaluative hidden tests will be run when you submit your code to the Gradescope autograder via the online student portal, and will provide feedback on how many points you have earned.

For debugging purposes, you can run a single unit test locally. For example, you can run the test case `3a-0-basic` using the following terminal command within the `src/` subdirectory:

```
$ python grader.py 3a-0-basic
```

Before beginning this course, please walk through the [Anaconda Setup for XCS Courses](#) to familiarize yourself with the coding environment. Use the env defined in `src/environment.yml` to run your code. This is the same environment used by the online autograder.

Test Cases

The autograder is a thin wrapper over the python `unittest` framework. It can be run either locally (on your computer) or remotely (on SCPD servers). The following description demonstrates what test results will look like for both local and remote execution. For the sake of example, we will consider two generic tests: `1a-0-basic` and `1a-1-hidden`.

Local Execution - Hidden Tests

All hidden tests rely on files that are not provided to students. Therefore, the tests can only be run remotely. When a hidden test like `1a-1-hidden` is executed locally, it will produce the following result:

```
----- START 1a-1-hidden: Test multiple instances of the same word in a sentence.
----- END 1a-1-hidden [took 0:00:00.011989 (max allowed 1 seconds), ???/3 points] (hidden test ungraded)
```

Local Execution - Basic Tests

When a basic test like `1a-0-basic` passes locally, the autograder will indicate success:

```
----- START 1a-0-basic: Basic test case.
----- END 1a-0-basic [took 0:00:00.000062 (max allowed 1 seconds), 2/2 points]
```

When a basic test like `1a-0-basic` fails locally, the error is printed to the terminal, along with a stack trace indicating where the error occurred:

```
----- START 1a-0-basic: Basic test case.
<class 'AssertionError'>
{'a': 2, 'b': 1} != None ← This error caused the test to fail.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a") ← In this case, start your debugging
                                           in line 23 of grader.py.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
----- END 1a-0-basic [took 0:00:00.003809 (max allowed 1 seconds), 0/2 points]
```

Remote Execution

Basic and hidden tests are treated the same by the remote autograder. Here are screenshots of failed basic and hidden tests. Notice that the same information (error and stack trace) is provided as the in local autograder, now for both basic and hidden tests.

1a-0-basic) Basic test case. (0.0/2.0)

```
<class 'AssertionError': {'a': 2, 'b': 1} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a"))
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

Just like in the local autograder, this error caused the test to fail.

Just like in the local autograder, start your debugging in line 23 of grader.py.

1a-1-hidden) Test multiple instances of the same word in a sentence. (0.0/3.0)

```
<class 'AssertionError': {'a': 23, 'ab': 22, 'aa': 24, 'c': 16, 'b': 15} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 31, in test_1
    self.compare_with_solution_or_wait(submission, 'extractWordFeatures', lambda f: f(sentence))
File "/autograder/source/graderUtil.py", line 183, in compare_with_solution_or_wait
    self.assertEqual(ans1, ans2)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

This error caused the test to fail.

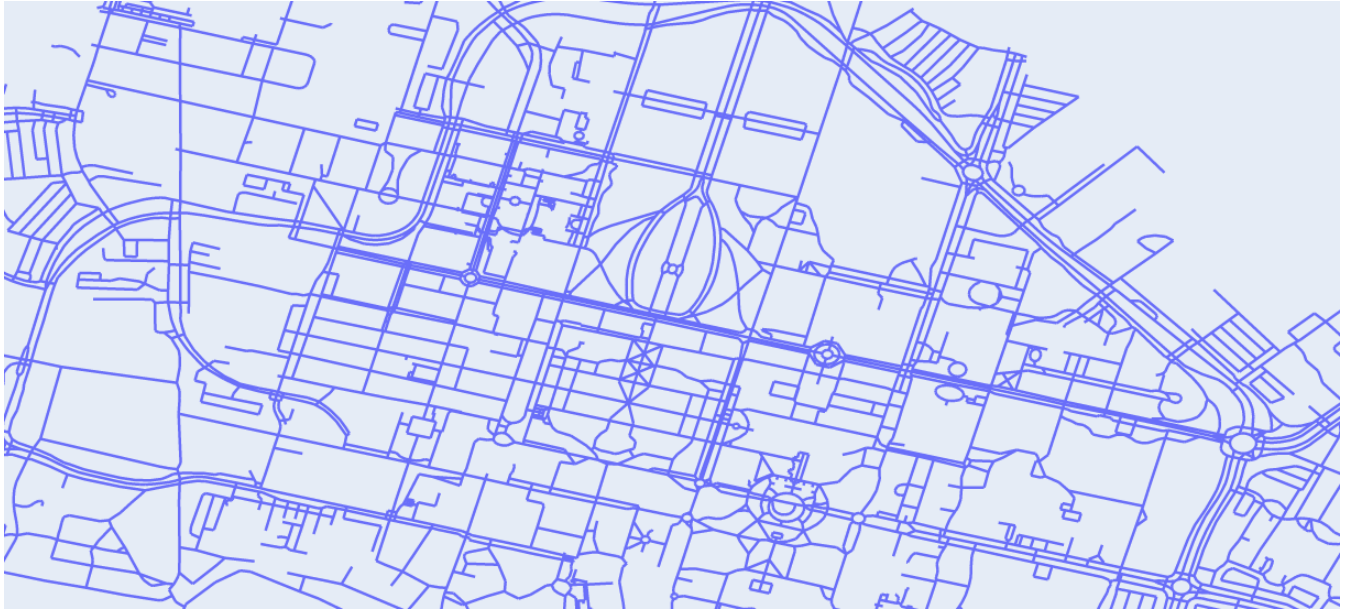
Start your debugging in line 31 of grader.py.

Finally, here is what it looks like when basic and hidden tests pass in the remote autograder.

1a-0-basic) Basic test case. (2.0/2.0)

1a-1-hidden) Test multiple instances of the same word in a sentence. (3.0/3.0)

Introduction



In route planning, the objective is to find the best way to get from point A to point B (think Google Maps). In this homework, we will build on top of the classic shortest path problem to allow for more powerful queries. For example, not only will you be able to explicitly ask for the shortest path from the Gates Building to Coupa Cafe at Green Library, but you can ask for the shortest path from Gates back to your dorm, stopping by the package center, gym, and the dining hall (in any order!) along the way.

We will assume that we have a *map* of a city (e.g., Stanford) consisting of a set of *locations*. Each location has:

1. a unique label (e.g., 6608996258),
2. a (latitude, longitude) pair specifying where the location is (e.g., 37.4299866, -122.175519), and
3. a set of *tags* which describes the type of location (e.g., amenity=food).

There are a set of *connections* between pairs of locations; each connection has a *distance* (in meters), and can be traversed in both directions (if the distance from A to B is 100 meters, then the distance from B to A is also 100 meters).

There are two city maps that you'll be working with: a grid map (`createGridMap`) and a map of Stanford (`createStanfordMap`), which is derived from [Open Street Maps](#). Please note that you do not need to know the Stanford campus in order to answer the questions below! We have also included instructions on how to create your own maps in `README.md`.

Important: On Windows some of the commands will lead to the python process not exiting and hanging. If you experience this behaviour, simply kill the process as the results will not be affected!

1. Grid City

Consider an infinite city consisting of locations (x, y) where x, y are integers. From each location (x, y) , one can go east, west, north, or south. You start at $(0, 0)$ and want to go to (m, n) , where $m, n \geq 0$. We can define the following search problem to capture this:

- $s_{\text{start}} = (0, 0)$
- $\text{Actions}(s) = \{(+1, 0), (-1, 0), (0, +1), (0, -1)\}$
- $\text{Succ}(s, a) = s + a$
- $\text{Cost}((x, y), a) = 1 + \max(x, 0)$ (it is more expensive as x increases)
- $\text{IsEnd}(s) = \mathbf{1}[s = (m, n)]$

(a) [1 point (Written)]

What is the minimum cost of reaching location (m, n) (note that $m, n \geq 0$) starting from location $(0, 0)$ in the above city? Describe one possible path achieving the minimum cost. Is it unique (i.e., are there multiple paths that achieve the minimum cost)?

What we expect: Provide an expression for the minimum cost. In addition, using 1 - 2 sentences describe one possible minimum cost path and state whether it is unique.

The minimum cost of reaching (m, n) from $(0, 0)$ is

$$\frac{m(m+1)}{2} + n.$$

The cost only depends on the x being left, so take all n north steps while still at $x = 0$, where each costs 1 and they total n , then the m east steps leave from $x = 0, 1, \dots, m-1$ in turn and cost $1, 2, \dots, m$; writing that series forwards and backwards pairs it into m columns of $m+1$, so it sums to $m(m+1)/2$. One path achieving the minimum is to take all n steps north from the origin first and then all m steps east, and it is the only one, since any north step taken after moving east costs more than 1 because it leaves from $x > 0$, and a detour into negative x adds at least one step west and one step back east, each costing at least 1, without making any other step cheaper.

(b) [1.50 points (Written)] How will Uniform Cost Search (UCS) behave on this problem? Mark the following as true or false:

- UCS will never terminate because the number of states is infinite.
- UCS will return the minimum cost path and explore only locations between $(0, 0)$ and (m, n) ; that is, (x, y) such that $0 \leq x \leq m$ and $0 \leq y \leq n$.
- UCS will return the minimum cost path and explore only locations whose past costs are less than the minimum cost from $(0, 0)$ to (m, n) .

What we expect: T/F for each subpart.

- False. The states go on forever, but UCS takes them in cost order and every step costs at least 1, so anything cheaper than the goal is only so many steps out from the origin. There are only so many of those, so it gets to the goal and stops.
- False. It does give back the minimum cost path, but it looks outside the box on the way. $(-1, 0)$ and $(0, -1)$ both cost 1 to get to, which is cheaper than most goals, so they get processed even though they sit outside $0 \leq x \leq m, 0 \leq y \leq n$.
- False. It returns the minimum cost path, but it can also look at places costing exactly the same as the minimum rather than less. When things tie, the queue decides who goes first. For the goal $(1, 0)$, which costs 1, the states $(-1, 0)$, $(0, -1)$ and $(0, 1)$ also cost 1 and come out before it.

(c) [1.50 points (Written)] Now consider running UCS on an arbitrary graph. Mark the following as true or false:

- If you add a connection between two locations, the minimum distance cannot go up.

- ii. If you make the cost of an action from some state small enough (possibly negative), that action will show up in the minimum cost path.
- iii. If you increase the cost of each action by 1, the minimum cost path does not change (even though its cost does).

What we expect: T/F for each subpart.

- i. True. Adding a connection only gives you more paths, it never takes one away, so the minimum can drop or stay where it is but it cannot go up.
- ii. False. An action coming out of a state the start cannot get to is not on any path from start to goal, however cheap it is. Say $A \xrightarrow{5} C$ is the only way from start A to goal C , and $D \xrightarrow{-1000} E$ sits off on its own, then that cheap action never shows up. Same if it leads into a dead end.
- iii. False. Adding 1 to every action charges a path by how many edges it has, so the longer ones get hit harder and the order can flip. Route 1 is $A \xrightarrow{10} Z$ with one edge, route 2 is $A \xrightarrow{2} B \xrightarrow{2} C \xrightarrow{2} D \xrightarrow{2} Z$ with four. Before, route 1 costs 10 and route 2 costs 8, so route 2 wins. After, route 1 costs 11 and route 2 costs 12, so route 1 wins.

2. Finding Shortest Paths

We first start out with the problem of finding the shortest path from a *start location* (e.g., the Gates Building) to some end location. In Google Maps, you can only specify a specific end location (e.g., Coupa Cafe at Green Library).

In this problem, we want to give the user the flexibility of specifying *multiple* possible end locations by specifying a set of "tags" (e.g., so you can say that you want to go to *any* place with food versus a specific location like Tressider).

(a) [8 points (Coding)]

Implement `ShortestPathProblem` so that given a `startLocation` and `endTag`, the minimum cost path corresponds to the shortest path from `startLocation` to *any* location that has the `endTag`.

In particular, you need to implement `startState()`, `isEnd(state)`, `successorsAndCosts(state)`.

Recall the separation between search problem (modeling) and search algorithm (inference). You should focus on modeling (defining the `ShortestPathProblem`); the default search algorithm, `UniformCostSearch` (UCS), is implemented for you in `util.py`.

What we expect: An implementation of the `ShortestPathProblem` class.

(b) [1 point (Written)]

Run `python mapUtil.py > readableStanfordMap.txt` to write a (long-ish) file of the possible locations on the Stanford map along with their tags. Each tag is a `[key]=[value]`. Here are some examples of keys:

- **landmark:** Hand-defined landmarks (from `src/data/stanford-landmarks.json`)
- **amenity:** Various amenity types (e.g., "park", "food")
- **parking:** Assorted parking options (e.g., "underground")

Choose a starting location and end tag (perhaps that's relevant to your daily life) and implement `getStanfordShortestPathProblem()` to create a search problem. Then, run `python grader.py 2b-0-helper` to generate `path.json`. Once generated, run `python visualization.py` to visualize it (opens in browser). Try different start locations and end tags. Pick two settings corresponding to the following:

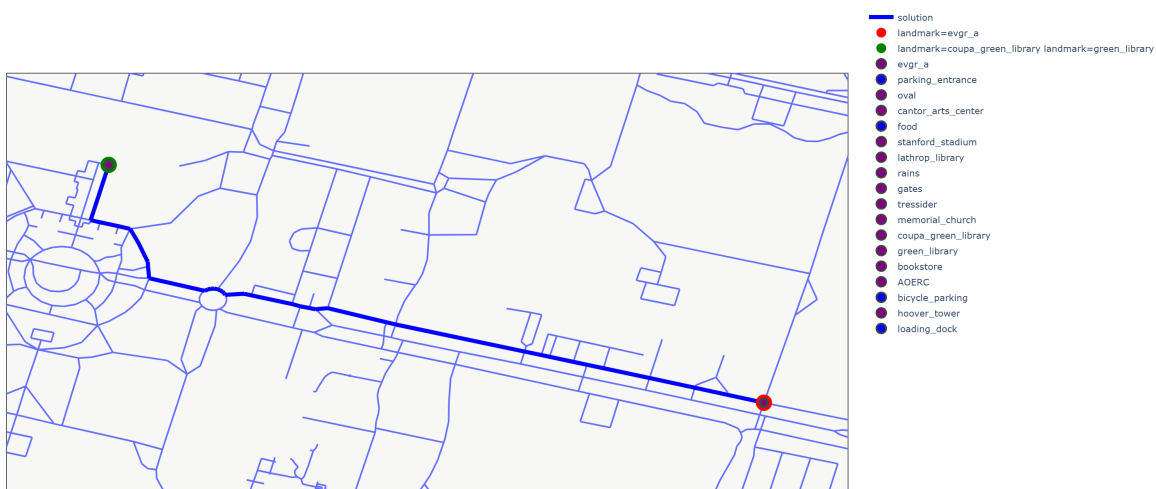
- A start location and end tag that produced new insight into traveling around campus. Describe whether the system was useful.
- A start location and end tag where the minimum cost path found isn't desirable. Is this due to incorrect modeling assumptions? Explain.

You should feel free to add new landmarks and if you are not at Stanford, follow the instructions in the `README.md` to use your own map and landmarks.

What we expect: A screenshot of the visualization of your two solutions as well as i) one or more sentences describing something interesting you've learned about traveling (around campus, or elsewhere) and ii) something incorrect about either the map or modeling assumptions (such a landmark being out of place, etc.).

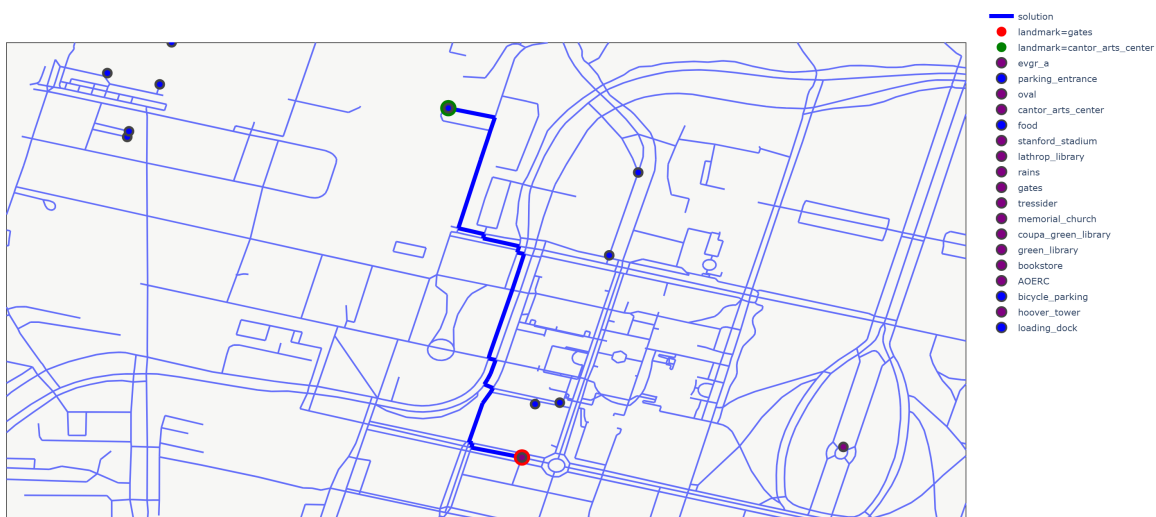
The first route start at EVGR-A with end tag `landmark=green_library`, which comes out around 606 m over 38 locations.

2b (useful): EVGR-A to Green Library



The second route start at Gates with end tag amenity=food, about 659 m, and it finishes at the Cantor Arts Center.

2b (undesirable): Gates to the nearest amenity=food



What I got out of it is that the cost is just distance, so anything that actually slows a walk down is free as far as the search is concerned. My EVGR-A route is around 606 m and goes through a couple of marked crossings, while the Gates one is roughly the same length but goes through about ten of them plus a bus stop, so the second walk takes a fair bit longer in practice and the search has no way to tell them apart. The locations are also spaced pretty unevenly, anywhere from under 2 m to nearly 100 m apart on the same path, so the number of steps does not say much about how far it is.

What seems wrong is that asking Gates for the nearest amenity=food sends to the Cantor Arts Center, while Tressider, which is where we would actually go, is further out and never gets picked. There are only a handful of

locations on the map with `amenity=food` at all, and a couple of them are the art museum and the football stadium, which happens because the tag is sitting on the landmark's own node, so one cafe somewhere inside a big building ends up labelling the whole thing as food. The assumption that breaks is that a tag on one node stands for the whole place, and that somewhere you can walk to is somewhere that is open.

(c) [1 point (Written)]

Your system now allows anyone to find the shortest path between any pair of locations on campus. By shortening their travel distance, it promises to optimize travel efficiency. But, there might be issues when a large fraction of the population start using your system to plan their routes.

In particular, what *negative externalities* might result from this system being widely deployed? Please refer to these brief articles for inspiration: "[Why Traffic Apps Make Congestion Worse](#)", "[The Perfect Selfishness of Mapping Apps](#)" as well as the module on Externalities ([video](#) and [pdf](#)).

Discuss the impact of these externalities on (1) users of your system and (2) non-users. Remember that these sorts of problems often arise from the mismatch between the real world and one's model of it. What are potential ways you could reduce this mismatch?

What we expect: Provide 3-5 sentences describing (a) two externalities (one for users of the route-planning system and one for other people/non-users of the system), as well as (b) at least one potential solution to reduce the impact of these problems.

If everyone plans with the same system, the users all get sent down the same shortest path, and the congestion that creates never shows up in the cost, because the cost is only distance. The people who do not use it are hit whenever their own route intersects that optimal path, so quiet side streets that were never built for through traffic end up carrying it, and they get none of the benefit since they never opted in. The mismatch is that the model prices an edge as a fixed number when the real price depends on how many people are on it at the time, and the system has no way of knowing that its own advice is what put them there. One way to reduce that is to put actual speed into the cost rather than distance alone, so a road that is slow or already saturated costs more and the system spreads people out instead of funnelling them all onto one route.

3. Finding Shortest Paths with Unordered Waypoints

Let's introduce an even more powerful feature: unordered waypoints! In Google Maps, you can specify an ordered sequence of waypoints that a path must go through – for example, going from point A to point X to point Y to point B, where [X, Y] are "waypoints" (such as a gas station or a friend's house).

However, we want to consider the case where the waypoints are *unordered*: X, Y, so that both $A \rightarrow X \rightarrow Y \rightarrow B$ and $A \rightarrow Y \rightarrow X \rightarrow B$ are allowed. Moreover, X, Y, and B are each specified by a tag like in Problem 2 (e.g., `amenity=food`)

This is a neat feature if you think about your day-to-day life; you might be on your way home after a long day, but need to stop by the package center, Tressider to grab a bite of food, and the bookstore to buy some notebooks. Having the ability to get a short, quick path that hits all these stops might be really convenient (rather than searching over the various waypoint orderings yourself).

(a) [11 points (Coding)]

Implement `WaypointsShortestPathProblem` so that given a `startLocation`, set of `waypointTags`, and an `endTag`, the minimum cost path corresponds to the shortest path from `startLocation` to a location with the `endTag`, such that all of `waypointTags` are covered by some location in the path.

Note that a single location can be used to satisfy multiple tags.

Like in Problem 2, you need to implement `startState()`, `isEnd(state)`, and `successorsAndCosts(state)`.

There are many ways to implement this search problem, so you should think carefully about how to design your `State`. We want to optimize for a compact state space so that search is as efficient as possible.

What we expect: An implementation of the `WaypointsShortestPathProblem` class. To get full credit, your implementation must have the right asymptotic dependence on the number of locations and waypoints. Note that your code will timeout if you do this incorrectly.

(b) [1.50 points (Written)]

If there are n locations and k waypoint tags, what is the maximum number of states that UCS could visit?

What we expect: A mathematical expression that depends on n and k , with a brief explanation justifying it.

The maximum number of states UCS could visit is

$$n \cdot 2^k.$$

A state is a location together with the set of waypoint tags covered so far, so there are n choices for the location, and for the memory each of the k tags is independently either covered or not, which gives 2^k possible sets. Multiplying the two independent choices gives $n \cdot 2^k$.

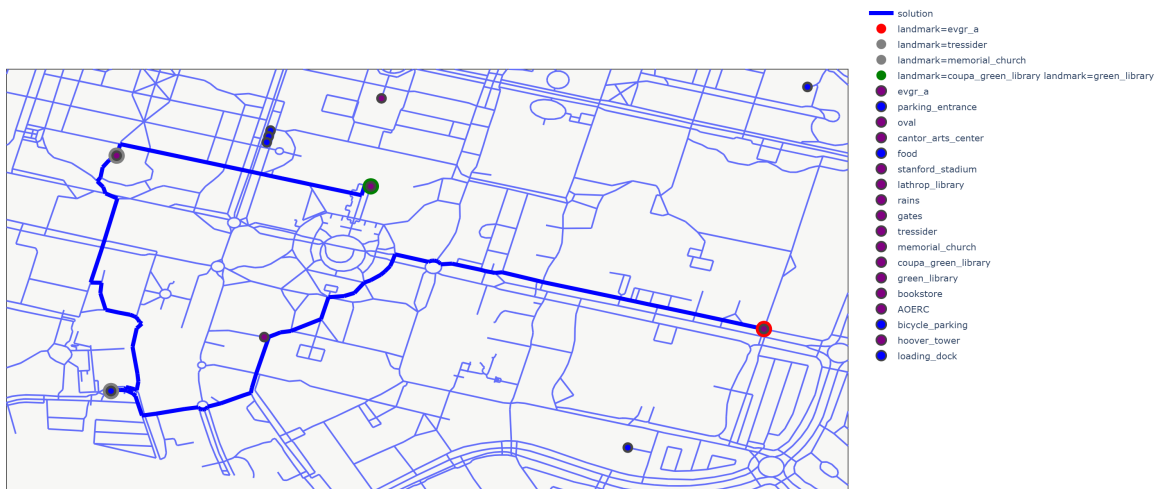
(c) [2 points (Written)]

Choose a starting location, set of waypoint tags, and an end tag (perhaps that captures an interesting route planning problem relevant to you), and implement `getStanfordWaypointsShortestPathProblem()` to create a search problem. Then, similar to Problem 1b, run `python grader.py 3c-0-helper` to generate `path.json`. Once generated, run `python visualization.py` to visualize it (opens in browser).

What we expect: A screenshot of the visualized path with a list of the waypoint tags you selected. In addition, provide one or more sentences describing unexpected or interesting features of the selected path. Does it match your expectations? What does this path fail to capture that might be important?

The waypoint tags are `landmark=memorial_church` and `landmark=tressider`, starting at `EVGR-A` and ending at `landmark=green_library`. It comes out around 1800 m over 130 locations.

3c: EVGR-A, via Memorial Church and Tressider, to Green Library



The order was not what I expected. I listed the church first but it went to Tressider first, then the church, then the library. That is the point of the waypoints being unordered, I do not have to pick the order myself, and my order would have cost more. The cost was about what I expected though. The same trip with no waypoints is around 600 m, so two stops made it roughly three times longer, which makes sense when the stops sit off to the side.

What it misses is everything that is not distance. It does not know Tressider closes in the evening, or that I would want to eat before the library and not after. It also only goes one way, so getting home is not counted at all.

- (d) [1.50 points (Written)] Ride sharing companies use route-finding systems similar to the one you built in this problem to route their drivers. However, these systems have been criticized for using exploitative behavioral nudges. For example, a [New York Times](#) article reported how one app uses “forward dispatch,” a feature that dispatches a new ride to a driver before the current one ends, to constantly keep drivers busy. This makes it more difficult for drivers to take breaks.

We want you to discuss how these companies could implement better labor practices, finding a way to use the unordering waypoints feature to balance company goals with the protection of driver interests. Specifically, we want you to think about ways your new unordered waypoints feature could help drivers in their working experience.

What waypoints could you include from one drop-off location to the next pick-up location to improve the driver experience? What information about drivers would you need to determine appropriate waypoints?

Note, however, that the unordered waypoints feature is an example of a dual use technology. For more, refer to the module on dual use technologies to help answer this question ([video](#) and [pdf](#)). Thus, we also want you to answer: What are some ways that a company could use this waypoints feature to increase profits at the expense of driver health? What are any downsides in collecting the appropriate information you need to recommend the waypoints?

What we expect: Provide 3-5 sentences describing (a) the waypoints you would include to improve the driver experience, (b) at least one dimension of drivers’ identity that could inform selection of waypoints, and (c) a potential negative use of the waypoints feature that a company could use to increase profits while harming the driver experience or driver health, or downsides in collecting the appropriate information needed to recommend the waypoints.

Between a drop-off and the next pick-up I would insert a rest waypoint, tagged for a restroom, food, or somewhere to park for ten minutes, in the same way problem 4c joins a fake node to every end location at zero cost so the search finds whichever one is nearest. Here the cost would be small rather than zero, so a break is cheap enough that the route takes one but not so free that the driver is sent miles out of the way for it. To pick sensible waypoints

I would want the driver's age and years of driving experience, since a new driver and a driver in their seventies do not need the same break pattern or the same length of shift. Those same two fields are exactly what a company could turn against them: knowing a driver's age lets dispatch quietly stop sending work to older drivers, which is age discrimination dressed up as routing, and it never has to be written down anywhere because it only shows up as fewer jobs.

4. Speeding up Search with A*

In this problem, we will explore how to speed up search by reducing the number of states that need to be expanded using various A* heuristics.

(a) [4 points (Coding)]

In this problem, you will implement A* to speed up the search. Recall that in class, A* is just UCS with a different cost function (a new search problem), so we will implement A* via a *reduction* to UCS.

In particular, you should implement `aStarReduction()` which takes a `search problem` and a `heuristic` as input and returns a new search problem.

What we expect: An implementation of the `NewSearchProblem` class in the `aStarReduction(problem, heuristic)` function. As in prior problems, you need to implement `startState()`, `isEnd(state)`, and `successorsAndCosts(state)`.

(b) [6 points (Coding)]

We are now ready to speed up search by simply implementing various heuristics. First, implement `StraightLineHeuristic` for Problem 2, which returns the straight line distance from a state to any location with the end tag. Note: you might want to precompute some things so that evaluating the heuristic is fast.

What we expect: An implementation of the `StraightLineHeuristic` class.

(c) [7 points (Coding)]

Next, implement `NoWaypointsHeuristic` for Problem 3, which returns the minimum cost path from a state to any location with the end tag but ignoring all the waypoints (essentially the solution to Problem 2, so you can reuse that if you'd like). Note: you might want to precompute some things so that evaluating the heuristic is fast. Helpful comments are provided in the code.

What we expect: An implementation of the `NoWaypointsHeuristic` class.

5. AI Tutor

In this problem, you'll participate in an interactive tutoring session with an AI chatbot. The sessions focus on dynamic programming and beam search. By the end, you should be able to explain and apply the basic principles of both techniques, understanding how they help AI systems make efficient and informed decisions.

We recommend using the below prompt template:

Prompt Template

INSERT PROMPT HERE. Keep the session around 15–20 minutes and interactive.

Guidelines

- Adjust difficulty: if I miss 2 in a row → slow + simpler; if I get 2 quickly → a bit harder or add a twist.
- Use tiny, deterministic examples (small numbers, short tables, brief ASCII diagrams, code blocks). Show actual *run-throughs* (step-by-step states) if helps.
- Use Code if it clarifies: ≤ 2 Python/NumPy blocks, each ≤ 20 lines; ideally no imports beyond `import numpy as np`.
- Do NOT solve graded work. If I paste any, refuse and make a similar practice item.
- Be CONCISE. Don't make notation complicated/unnecessary. Focus on intuitions. Code blocks should be short with simple variable names.
- I may interrupt and ask to start somewhere else. Simply adjust and adapt from there.

Suggested flow (flexible; use judgment)

- Start by explaining the topic in a few simple, but technically accurate sentences (explain core idea (≤ 6 sentences): intuition + minimal notation + why it matters + one common pitfall). Use basic notations and visualizations if necessary (CS221 requires basic math/programming backgrounds). For topics about programming, OK to start with concise code blocks.
- Quick check-in (≤ 1 –2 min): ask 2 short questions about prior exposure and specific goal.
- Exercises: use a tiny instance and show intermediate states. Give exercises that build on each other. OK to give multiple exercises at the same time.
 - e.g., for graph search: a 4–5 node weighted graph; show the frontier/priority-queue after each of 2–3 steps and which edges relax.
 - for DP/MDP/RL/Logic/ML, do an equivalently small, concrete step-by-step trace.
 - If code clarifies, include one concise NumPy block.
- 3 quick checks: increasing difficulty; give a hint first; reveal answers only after I try or ask.
- Assessment and feedback: Give me feedback on my performance - what I understood well, what needs improvement, and specific concepts to review further.
- Session reflection: Ask me for feedback on how the tutoring session went - what worked well, what could be improved, and if the pacing/difficulty was appropriate.
- Recap: a 60-second summary

(a) [0.50 points (Written)] Practice Dynamic Programming with an AI tutor!

Use an AI chatbot (e.g., ChatGPT, Claude, Gemini).

For the prompt, use **"Teach me basic Dynamic Programming interactively"**. We recommend using the template provided above.

What we expect: Provide a link to the chat session transcript with the AI tutor. The session should be around 15–20 minutes and interactive!

Interactive tutoring session on Dynamic Programming:

<https://claude.ai/share/96fd17a6-37a0-4650-a6fc-3c82c68c815b>

(b) **[0.50 points (Written)] Practice Beam Search with an AI tutor!**

Use an AI chatbot (e.g., ChatGPT, Claude, Gemini).

For the prompt, use **"Teach me basic Beam Search interactively"**. We recommend using the template provided above.

What we expect: Provide a link to the chat session transcript with the AI tutor. The session should be around 15–20 minutes and interactive!

Interactive tutoring session on Beam Search:

<https://claude.ai/share/c969fba9-4932-4670-9952-aa6afa912e3a>

6. Product Ethics Exploration: Documentation, Marketing, and Customer Support

You will choose a modern AI product to analyze throughout the assignments and gain initial access to it. Please note that the questions regarding your chosen product are interconnected and will progressively build upon each other from one assignment to the next.

Suggested resources:

- Pick an AI-powered product or tool (not just a raw LLM chatbot) that is accessible to you. Good candidates include: Cursor, Nano Banana, ElevenLabs, Perplexity, Claude Code, etc. Free trials or student-accessible tools are recommended.
- The product's official website and onboarding/getting started pages
- Privacy policy, Terms of Service, Acceptable Use Policy
- App store listings, GitHub documentation or READMEs (if open-source or developer-targeted)
- Recent news coverage (e.g., TechCrunch, The Verge, Wired, NYT technology section)
- Reddit communities, Hacker News, or user forums for first impressions and experiences

In this part: You will examine how the company communicates with users through marketing, documentation, and support channels.

Why it's important: The gap between marketing promises and actual capabilities, as well as the quality of user support, directly impacts user trust and the product's real-world effectiveness.

Suggested resources:

- Company marketing pages, ads, YouTube demos
- Social media promotions (LinkedIn, TikTok, Instagram, X)
- Reddit, StackOverflow, Discord, or unofficial user communities
- Product FAQ, Help Center
- Customer service emails/chat logs you submit

(a) [0.50 points (Written)] Marketing Channels and Discovery

- How is the product marketed and how do users discover it? Describe the channels and messaging the company uses.

What we expect: In 4-6 sentences, provide a summary of the main marketing channels and discovery avenues for the product, with links, cited examples, and/or screenshots.

My product is Claude Code, carried over from Assignment 1. It is marketed through non-consumer channels rather than advertising: new releases go out through Anthropic's own newsroom and social media and get picked up by news outlets from there, and the product page at claude.com/product/claude-code carries the pitch and the install line in the same place. Anthropic also runs its own developer conference, Code with Claude, which in 2026 ran in San Francisco, London and Tokyo. The other way in is work, since it is sold to businesses on Team and Enterprise plans (claude.com/pricing), so plenty of developers meet it because their employer signed an agreement rather than because they went looking. Past that it is mostly word of mouth in developer communities, and the brand level advertising, like the Super Bowl spots in February 2026, is for Claude as a whole and never mentions Claude Code.

(b) [0.50 points (Written)] Marketing Strategy and Mission Alignment

- Summarize what you believe the company's **core mission** is, based on official statements (About page, press releases, etc.).
- Discuss whether the product's marketing strategy and messaging **align or conflict with the company's mission**. In your view, does the marketing sensationalize or misrepresent what the product can do, or is it fair and accurate? Provide rationale and cite examples (ads, taglines, posts, etc.).

What we expect: In 4-6 sentences, clearly articulate whether the company's marketing supports, contradicts, or sensationalizes its stated mission and product capabilities, providing specific evidence.

Anthropic's mission on its company page is "Anthropic is an AI safety and research company. We build reliable, interpretable, and steerable AI systems" (anthropic.com/company), and I think the marketing lines up with it rather

than overselling. The Life Sciences Verification Program gives verified researchers access to biology work the general models block (anthropic.com/news/life-sciences-verification-program), which is the steerable part of that mission shipped as a product. The September 2026 threat intelligence report goes the other way and documents real misuse of Claude across seven categories (anthropic.com/threat-intelligence-report-september-2026), which is not something you publish if you are exaggerating what your models can do. So I would call it fair rather than sensationalised.

(c) [0.50 points (Written)] **Feedback, Help, and Customer Support**

- How can users contact the company or get support? Describe the official support channels, any community-based help resources, and your overall impression of how easy it is to get assistance.

What we expect: In 2-3 sentences, summarize the avenues available for support and community help, and comment on the effectiveness of official and unofficial documentation resources.

Support is written only. The help page states plainly that “We don’t offer phone support” and that “Support is provided in writing through the support messenger and email”, with an AI bot called Fin answering first and a human product support team reachable only on Pro, Max, Team and Enterprise, while free users get the documentation and the bot (support.claude.com). The process is asynchronous by design, so a specialist “investigates and replies in the same conversation” with response times varying by plan and severity, and there is no realtime phone or video route at all. For Claude Code specifically the community channels carry more of the load: the official documentation at code.claude.com is good on features, but the /bug command and the GitHub issue tracker at anthropics/claude-code, which currently holds over five thousand open issues, are where you find out whether a problem is yours or everyone’s, alongside the Claude Developers Discord and the r/ClaudeAI subreddit.

(d) [0.50 points (Written)] **Submitting Customer Support Queries**

- Submit **two customer support queries** based on your personal experience using the product. These should reflect genuine questions or typical issues you encountered. List those queries here and reflect in 1-2 sentences how customer support responded to and handled your queries.

What we expect: Include two well-constructed sample support emails or chat messages, each clearly stating a question, problem, or request for help and, for each query, 1-2 sentences reflecting on how the company handled that query.

Both queries went through the support messenger at support.claude.com.

Query 1, sent about Claude Code:

I use Claude Code on a Max plan and I’d like to know whether it can be pointed at a different model endpoint. I know Bedrock and Vertex are supported. Is there any supported way to use an OpenAI-compatible or OpenRouter endpoint, and if not, is that something on the roadmap? I’m asking because I want to compare outputs on the same repo without changing tools.

The reply:

Claude Code can connect through LLM gateways that expose a supported API format, but we don’t support routing Claude Code to non-Claude models through any gateway.

While you can configure Claude Code to use different Claude model endpoints (like Bedrock, Vertex, Microsoft Foundry, or through custom ANTHROPIC_BASE_URL settings), the gateway compatibility is specifically for accessing Claude models, not for switching to other model providers like OpenAI or OpenRouter.

For comparing outputs on the same repo, you’d need to use separate tools for non-Claude models. We don’t have information about future roadmap plans for multi-model support in Claude Code.

Query 2, sent about Claude itself:

I’m using Claude to learn the maths for a machine learning course, and I want it to tutor me rather than solve things for me. The problem I keep hitting is that when I’m deriving something, Claude already knows the final form and reasons backwards toward it, which looks like a derivation but doesn’t help, because the answer is the one thing I don’t have. Telling it to stop works for a few messages and then it drifts back. Where is the best place to put an instruction like that so it holds for a whole session, Project instructions, a custom style, or something else? And is there a recommended way to phrase it?

The reply:

For persistent tutoring instructions, Project instructions are your best option. They apply to all chats within that project, so you can keep your ML course work in one project with the tutoring behavior locked in. Here's how to phrase it effectively. In your project instructions, try something like: "Act as a tutor, not a solver. When I'm deriving equations or working through problems, guide me with hints and questions rather than showing the final form or working backwards from the answer. Let me discover the solution step-by-step. If I get stuck, ask what I've tried or suggest the next small step, but don't reveal the destination."

You could also add this to your account-wide Instructions for Claude (Settings → Instructions for Claude) if you want this tutoring approach across all your conversations, not just in one project. Project instructions work on top of your account instructions, so you can combine both, general preferences account-wide and specific tutoring rules in your ML project.

Both queries were returned through the online fin agent without human replies.

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset L^AT_EX solutions.

1.a

The minimum cost of reaching (m, n) from $(0, 0)$ is

$$\frac{m(m+1)}{2} + n.$$

The cost only depends on the x being left, so take all n north steps while still at $x = 0$, where each costs 1 and they total n , then the m east steps leave from $x = 0, 1, \dots, m-1$ in turn and cost $1, 2, \dots, m$; writing that series forwards and backwards pairs it into m columns of $m+1$, so it sums to $m(m+1)/2$. One path achieving the minimum is to take all n steps north from the origin first and then all m steps east, and it is the only one, since any north step taken after moving east costs more than 1 because it leaves from $x > 0$, and a detour into negative x adds at least one step west and one step back east, each costing at least 1, without making any other step cheaper.

1.b

1. False. The states go on forever, but UCS takes them in cost order and every step costs at least 1, so anything cheaper than the goal is only so many steps out from the origin. There are only so many of those, so it gets to the goal and stops.
2. False. It does give back the minimum cost path, but it looks outside the box on the way. $(-1, 0)$ and $(0, -1)$ both cost 1 to get to, which is cheaper than most goals, so they get processed even though they sit outside $0 \leq x \leq m$, $0 \leq y \leq n$.
3. False. It returns the minimum cost path, but it can also look at places costing exactly the same as the minimum rather than less. When things tie, the queue decides who goes first. For the goal $(1, 0)$, which costs 1, the states $(-1, 0)$, $(0, -1)$ and $(0, 1)$ also cost 1 and come out before it.

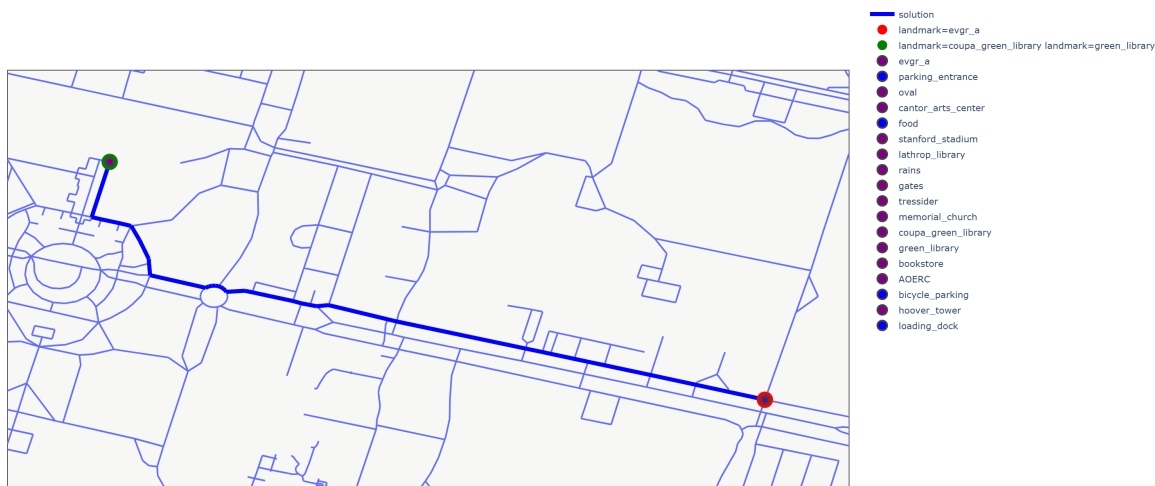
1.c

1. True. Adding a connection only gives you more paths, it never takes one away, so the minimum can drop or stay where it is but it cannot go up.
2. False. An action coming out of a state the start cannot get to is not on any path from start to goal, however cheap it is. Say $A \xrightarrow{5} C$ is the only way from start A to goal C , and $D \xrightarrow{-1000} E$ sits off on its own, then that cheap action never shows up. Same if it leads into a dead end.
3. False. Adding 1 to every action charges a path by how many edges it has, so the longer ones get hit harder and the order can flip. Route 1 is $A \xrightarrow{10} Z$ with one edge, route 2 is $A \xrightarrow{2} B \xrightarrow{2} C \xrightarrow{2} D \xrightarrow{2} Z$ with four. Before, route 1 costs 10 and route 2 costs 8, so route 2 wins. After, route 1 costs 11 and route 2 costs 12, so route 1 wins.

2.b

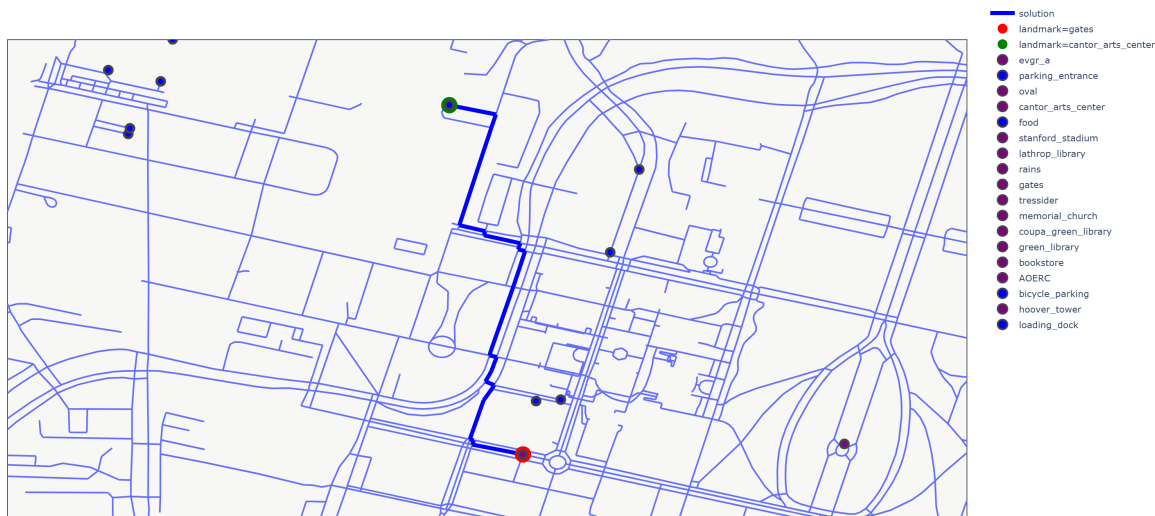
The first route start at EVGR-A with end tag `landmark=green_library`, which comes out around 606 m over 38 locations.

2b (useful): EVGR-A to Green Library



The second route start at Gates with end tag `amenity=food`, about 659 m, and it finishes at the Cantor Arts Center.

2b (undesirable): Gates to the nearest amenity=food



What I got out of it is that the cost is just distance, so anything that actually slows a walk down is free as far as the search is concerned. My EVGR-A route is around 606 m and goes through a couple of marked crossings, while the Gates one is roughly the same length but goes through about ten of them plus a bus stop, so the second walk takes a fair bit longer in practice and the search has no way to tell them apart. The locations are also spaced pretty unevenly, anywhere from under 2 m to nearly 100 m apart on the same path, so the number of steps does not say much about how far it is.

What seems wrong is that asking Gates for the nearest `amenity=food` sends to the Cantor Arts Center, while Tressider, which is where we would actually go, is further out and never gets picked. There are only a handful of locations on the map with `amenity=food` at all, and a couple of them are the art museum and the football stadium, which happens because the tag is sitting on the landmark's own node, so one cafe somewhere inside a big building ends up labelling the whole thing as food. The assumption that breaks is that a tag on one node stands for the whole place, and that somewhere you can walk to is somewhere that is open.

2.c

If everyone plans with the same system, the users all get sent down the same shortest path, and the congestion that creates never shows up in the cost, because the cost is only distance. The people who do not use it are hit whenever their own route intersects that optimal path, so quiet side streets that were never built for through traffic end up carrying it, and they get none of the benefit since they never opted in. The mismatch is that the model prices an edge as a fixed number when the real price depends on how many people are on it at the time, and the system has no way of knowing that its own advice is what put them there. One way to reduce that is to put actual speed into the cost rather than distance alone, so a road that is slow or already saturated costs more and the system spreads people out instead of funnelling them all onto one route.

3.b

The maximum number of states UCS could visit is

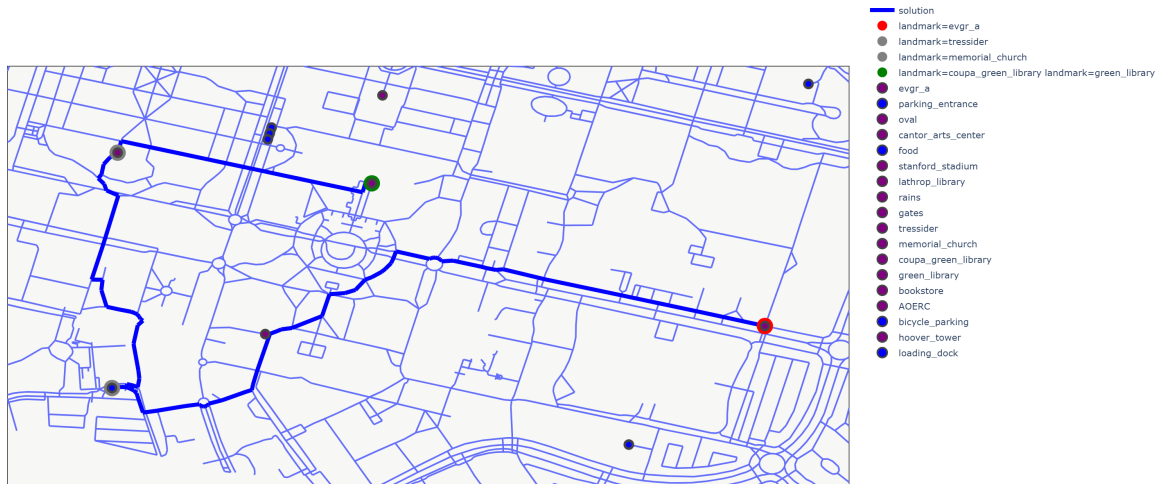
$$n \cdot 2^k.$$

A state is a location together with the set of waypoint tags covered so far, so there are n choices for the location, and for the memory each of the k tags is independently either covered or not, which gives 2^k possible sets. Multiplying the two independent choices gives $n \cdot 2^k$.

3.c

The waypoint tags are `landmark=memorial_church` and `landmark=tressider`, starting at `EVGR-A` and ending at `landmark=green_library`. It comes out around 1800 m over 130 locations.

3c: EVGR-A, via Memorial Church and Tressider, to Green Library



The order was not what I expected. I listed the church first but it went to Tressider first, then the church, then the library. That is the point of the waypoints being unordered, I do not have to pick the order myself, and my order would have cost more. The cost was about what I expected though. The same trip with no waypoints is around 600 m, so two stops made it roughly three times longer, which makes sense when the stops sit off to the side.

What it misses is everything that is not distance. It does not know Tressider closes in the evening, or that I would want to eat before the library and not after. It also only goes one way, so getting home is not counted at all.

3.d

Between a drop-off and the next pick-up I would insert a rest waypoint, tagged for a restroom, food, or somewhere to park for ten minutes, in the same way problem 4c joins a fake node to every end location at zero cost so the search finds whichever one is nearest. Here the cost would be small rather than zero, so a break is cheap enough that the route takes one but not so free that the driver is sent miles out of the way for it. To pick sensible waypoints I would want the driver's age and years of driving experience, since a new driver and a driver in their seventies do not need the same break pattern or the same length of shift. Those same two fields are exactly what a company could turn against them: knowing a driver's age lets dispatch quietly stop sending work to older drivers, which is age discrimination dressed up as routing, and it never has to be written down anywhere because it only shows up as fewer jobs.

5.a

Interactive tutoring session on Dynamic Programming:

<https://claude.ai/share/96fd17a6-37a0-4650-a6fc-3c82c68c815b>

5.b

Interactive tutoring session on Beam Search:

<https://claude.ai/share/c969fba9-4932-4670-9952-aa6afa912e3a>

6.a

My product is Claude Code, carried over from Assignment 1. It is marketed through non-consumer channels rather than advertising: new releases go out through Anthropic's own newsroom and social media and get picked up by news outlets from there, and the product page at claude.com/product/claude-code carries the pitch and the install line in the same place. Anthropic also runs its own developer conference, Code with Claude, which in 2026 ran in San Francisco, London and Tokyo. The other way in is work, since it is sold to businesses on Team and Enterprise plans (claude.com/pricing), so plenty of developers meet it because their employer signed an agreement rather than because they went looking. Past that it is mostly word of mouth in developer communities, and the brand level advertising, like the Super Bowl spots in February 2026, is for Claude as a whole and never mentions Claude Code.

6.b

Anthropic's mission on its company page is "Anthropic is an AI safety and research company. We build reliable, interpretable, and steerable AI systems" (anthropic.com/company), and I think the marketing lines up with it rather than overselling. The Life Sciences Verification Program gives verified researchers access to biology work the general models block (anthropic.com/news/life-sciences-verification-program), which is the steerable part of that mission shipped as a product. The September 2026 threat intelligence report goes the other way and documents real misuse of Claude across seven categories (anthropic.com/threat-intelligence-report-september-2026), which is not something you publish if you are exaggerating what your models can do. So I would call it fair rather than sensationalised.

6.c

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6.d

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